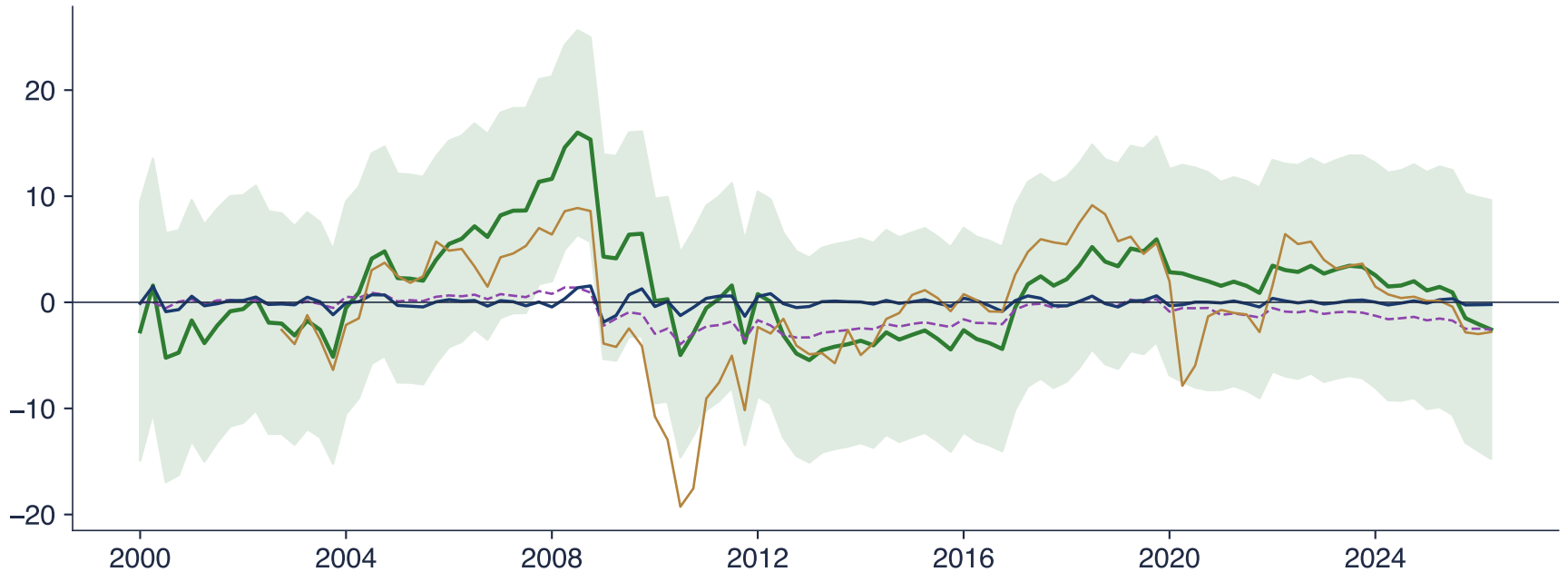


% of potential output



smooth-trend UC, smoothed gap

smooth-trend UC, filtered (one-sided) gap

Hamilton (2018) cycle

90% band

random-walk-trend UC (UC0), smoothed cycle